TWO COUNTEREXAMPLES TO A PACKING-COLORING PROBLEM OF MORTADA, EL ZEIN, AND AL HAJJAR

Abstract. Mortada, El Zein, and Al Hajjar asked whether every connected claw-free subcubic graph other than a specified graph $\mathcal{H}$ is $(1,1,3,3,4)$-packing colorable. We give two nonisomorphic counterexamples, both cubic of order 36. Each is the triangle expansion of a cubic graph of order 12, and its noncolorability follows from an elementary analysis of the possible supports of 4-packings.

1. Construction and statement

For a positive integer t, a *t-packing* in a graph is a set of vertices at pairwise distance greater than t. For $S = (s_1, \ldots, s_k)$, an *S-packing coloring* is a partition $V(G) = A_1 \sqcup \cdots \sqcup A_k$ in which A_i is an s_i-packing. Mortada, El Zein, and Al Hajjar asked whether every connected claw-free subcubic graph distinct from their graph $\mathcal{H}$ is $(1,1,3,3,4)$-packing colorable [1, Problem 1].

Let C be a cubic graph. Its *triangle expansion* $T(C)$ is obtained by replacing every $u \in V(C)$ with a triangle Δ_u, whose three vertices are assigned bijectively to the three edges incident with u, and by joining the assigned vertices corresponding to each edge of C.

We define two cubic cores. The graph C_1 has vertex set

$$\{r, s, q_0, q_1, x, y, a_r, a_s, a_0, b_r, b_s, b_0\}.$$

The set $\{q_0, q_1, x, y\}$ induces a diamond with missing edge xy; the sets

$$A = \{a_r, a_s, a_0\}, \qquad B = \{b_r, b_s, b_0\}$$

induce triangles; and the remaining edges are

$$rx,\ ra_r,\ rb_r,\ sy,\ sa_s,\ sb_s,\ a_0b_0.$$

The graph C_2 has vertices p_i, q_i, x_i, y_i for $i \in \mathbb{Z}/3\mathbb{Z}$. For each i, the set

$$D_i = \{p_i, q_i, x_i, y_i\}$$

induces a diamond with missing edge x_iy_i, and the remaining edges are

$$x_iy_{i+1} \qquad (i \in \mathbb{Z}/3\mathbb{Z}).$$

Put $G_1 = T(C_1)$ and $G_2 = T(C_2)$. For reference, graph6 encodings of C_1 and C_2 are `K?`CPagXAcAg` and `K?`CPagPagEG`, in which the third character of each is the ASCII grave accent, byte `60` hexadecimal.

2020 *Mathematics Subject Classification.* 05C15.

Key words and phrases. packing coloring, claw-free graph, cubic graph, counterexample.

Theorem 1. *The graphs G_1 and G_2 are nonisomorphic connected cubic claw-free graphs of order 36, and neither is $(1,1,3,3,4)$-packing colorable. Consequently, Problem 1 of Mortada, El Zein, and Al Hajjar has a negative answer.*

Both cores are connected and cubic of order 12. In a triangle expansion the external edges form a matching, so the replacement triangles are all the triangles. Hence $T(C)$ is connected, cubic, and claw-free, has order $3|V(C)|$, and determines C by contraction of its triangles. The core C_1 has four triangles and C_2 has six; therefore G_1 and G_2 are nonisomorphic. They have order 36, whereas $\mathcal{H}$ has order 12 [1, Figure 1]. It remains to prove noncolorability.

2. Distances and supports

Index the vertices of Δ_u by the three edges incident with u. The vertex indexed by an edge e will be called the *e-port* of Δ_u.

Lemma 2 (Port criterion)**.** *Let $u \neq v$ be vertices of a cubic graph C.*

(1) *If $uv \in E(C)$, then every vertex of Δ_u is at distance at most 3 from every vertex of Δ_v in $T(C)$.*
(2) *If $d_C(u,v) \geq 3$, then every vertex of Δ_u is at distance at least 5 from every vertex of Δ_v.*
(3) *Suppose $d_C(u,v) = 2$. Chosen ports in Δ_u and Δ_v are at distance greater than 4 if and only if, for every common neighbor w of u and v, neither chosen port is indexed by the edge toward w.*

Proof. A simple core path $u = u_0u_1\cdots u_k = v$ lifts between chosen endpoint ports to a path of length

$$2k - 1 + \varepsilon_u + \varepsilon_v,$$

where each ε is 0 or 1 according as the chosen endpoint port is or is not indexed by the corresponding end edge of the core path. Conversely, a shortest path in $T(C)$ that uses m external edges cannot traverse one immediately in both directions, and therefore uses at least $m-1$ intratriangle edges. Thus its length is at least $2m-1$. The first two assertions follow. If $d_C(u,v) = 2$, every path of length at most 4 must project to a path uwv and has length $3 + \varepsilon_u + \varepsilon_v$. This gives the last assertion. □

For $P \subseteq V(T(C))$, define its core support by

$$\operatorname{supp}_C(P) = \{u \in V(C) : P \cap \Delta_u \neq \varnothing\}.$$

Corollary 3. *If P is a t-packing in $T(C)$ with $t \geq 3$, then $\operatorname{supp}_C(P)$ is independent in C. In particular, $|P| \leq \alpha(C)$.*

Proof. The set P contains at most one vertex of each replacement triangle, and Lemma 2(1) excludes two adjacent core vertices from its support. □

Lemma 4. *One has $\alpha(C_1) = \alpha(C_2) = 4$.*

Proof. Let I be independent in C_1. If I contains neither r nor s, then it meets the diamond in at most two vertices and each of A, B in at most one. If it contains exactly one of r, s, say r, then the other vertices lie in the three cliques

$$\{q_0, q_1, y\}, \qquad \{a_s, a_0\}, \qquad \{b_s, b_0\},$$

so again $|I| \leq 4$. If it contains both r and s, only q_0, q_1, a_0, b_0 remain available, with edges q_0q_1 and a_0b_0. Thus $|I| \leq 4$, and equality is attained by $\{r, s, q_0, a_0\}$.

In C_2, an independent set meets a diamond D_i twice only in the nonedge $\{x_i, y_i\}$. Two distinct such pairs contain one of the linking edges x_iy_{i+1}, so at most one diamond is met twice. Hence $|I| \leq 2 + 1 + 1 = 4$, with equality attained by $\{p_0, p_1, x_2, y_2\}$. □

3. The two obstructions

Call p_i, q_i the internal vertices and x_i, y_i the external vertices of the diamond D_i.

Lemma 5. *Every 4-packing in G_2 has at most three vertices.*

Proof. Suppose that a 4-packing P has four vertices and put $I = \mathrm{supp}_{C_2}(P)$. By Corollary 3 and Lemma 4, I is independent of size four. It therefore meets one diamond, say D_i, in $\{x_i, y_i\}$ and each other diamond once. Since x_i and y_i have the two common neighbors p_i, q_i, the port criterion forces the selected ports in Δ_{x_i} and Δ_{y_i} to be the linking ports for x_iy_{i+1} and $x_{i-1}y_i$.

Independence excludes y_{i+1} and x_{i-1}. The two remaining external candidates, x_{i+1} and y_{i-1}, are adjacent. Hence one of the two remaining support vertices is internal. If it is internal in D_{i+1}, every port in its replacement triangle is at distance at most four from the selected x_iy_{i+1}-port: cross to $\Delta_{y_{i+1}}$, move within that triangle, cross to the internal vertex's triangle, and move within it. This contradicts the 4-packing condition. The case of an internal vertex in D_{i-1} is symmetric. □

Suppose that G_2 has a $(1, 1, 3, 3, 4)$-packing coloring with classes $A_1, \ldots, A_5$. Each replacement triangle contains at most one vertex of each of the independent sets A_1, A_2, and hence at least one vertex of $A_3 \cup A_4 \cup A_5$. Therefore

$$|A_3| + |A_4| + |A_5| \geq 12.$$

Corollary 3, Lemma 4, and Lemma 5 give the contradiction

$$|A_3| + |A_4| + |A_5| \leq 4 + 4 + 3 = 11.$$

Lemma 6. *Let P be a 4-packing of four vertices in G_1, and put $I = \mathrm{supp}_{C_1}(P)$. Then $C_1 - I$ contains a triangle.*

Proof. The support I is independent. We distinguish its intersection with $\{r, s\}$.

If $r, s \in I$, then I contains one of q_0, q_1 and one of a_0, b_0. If $a_0 \in I$, the triangle B lies in $C_1 - I$; if $b_0 \in I$, the triangle A does.

Suppose that $r \in I$ and $s \notin I$. The other three vertices of I consist of one from each of

$$\{q_0, q_1, y\}, \qquad \{a_s, a_0\}, \qquad \{b_s, b_0\}.$$

If $q_j \in I$, then the three other members of I are at core distance two from r through its three distinct neighbors x, a_r, b_r. Lemma 2(3) leaves no possible port at r, a contradiction. Hence $y \in I$. The pair a_0, b_0 is adjacent. The choice a_s, b_0 leaves no possible port at a_s, because r, y, b_0 are reached at core distance two through a_r, s, a_0, respectively; the choice a_0, b_s is symmetric. Thus

$$I = \{r, y, a_s, b_s\},$$

and $\{q_0, q_1, x\}$ is a triangle in $C_1 - I$. The case $s \in I$, $r \notin I$ is symmetric and leaves the triangle $\{q_0, q_1, y\}$.

Finally, suppose $r, s \notin I$. Then I contains x, y, one vertex of A, and one vertex of B. Since x, y have the two common neighbors q_0, q_1, the port criterion forces their selected ports to be the ports for xr and ys. The chosen vertices of A and B must consequently be a_0 and b_0, respectively, contradicting $a_0b_0 \in E(C_1)$. □

Suppose that G_1 has a $(1, 1, 3, 3, 4)$-packing coloring with classes $A_1, \dots, A_5$. As above,

$$|A_3| + |A_4| + |A_5| \geq 12.$$

Each of these three classes has size at most four, so equality holds throughout. Thus each has size four, and every replacement triangle contains exactly one vertex of $A_3 \cup A_4 \cup A_5$. Their core supports are therefore disjoint independent sets that partition $V(C_1)$. It follows that

$$C_1 - \mathrm{supp}_{C_1}(A_5)$$

is bipartite, with the other two supports as a bipartition. This contradicts Lemma 6. Theorem 1 follows.

Remark. No minimality claim is made. By Theorem 4 of [1], both examples are $(1, 1, 3, 3, 3)$-packing colorable. Thus they refute precisely the proposed strengthening in which the last distance parameter is increased from 3 to 4.

References

[1] M. Mortada, A. El Zein, and S. Al Hajjar, On $(1, 1, 2, 3)$- and $(1, 1, 3, 3, 3)$-packing colorings of claw-free subcubic graphs, arXiv:2608.02566v1 [math.CO], 3 August 2026.